

Ehud Netzer

HASMONEAN AND HERODIAN PALACES AT JERICHO

Final Reports of the 1973–1987 Excavations

Volume I: Stratigraphy and Architecture



Israel Exploration Society
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CHAPTER 2

AREA AC — THE SMALL SWIMMING POOLS

Area AC, located immediately to the west of the tell (see Plan 2), is a relatively level area measuring 75×40 m (see Ill. 72). Our attention was drawn to it by a number of pools discernible here (see Plan 12). One of them [Pool A(C)94], located at the center of this area, found expression as a real depression and there were signs of some excavation work here. Another, smaller pool [A(C)150], located in the area's southwestern corner, had been completely exposed before our study. A third pool [A(C)90], of roughly the same size as the first one, could be distinguished on the north of the area and lacked any signs of previous excavations. During our excavations, we revealed two other pools: Pool AC44, of more or less the same size as, and to the east of, Pool A(C)94; and a smaller pool [A(C)161], located to the north of Pool A(C)94.

Kelso and Baramki mention that they uncovered a small pool to the west of the tell, which was full of ceramic finds.¹ (These finds were so numerous that they linked them with the manufacture of ceramic vessels.) They also mention several architectural elements which were found in the same pool. It is difficult for us to determine whether they are referring to the large pool [A(C)94] at the center of Area AC, or to the small one [A(C)150] on the south. In any event, Plate 33 in their book deals only with the large pool; this map also features the third pool [A(C)90], to the north of the water conduit. On the basis of all these data, it appears that the pool mentioned by Kelso and Baramki is Pool A(C)94.

In the description of this area we shall first deal with the water systems passing through its northern part, most of which are the principal ones of the Buried Palace.



Ill. 72. General view of Area AC from the east.

WATER SYSTEMS WITHIN AREA AC

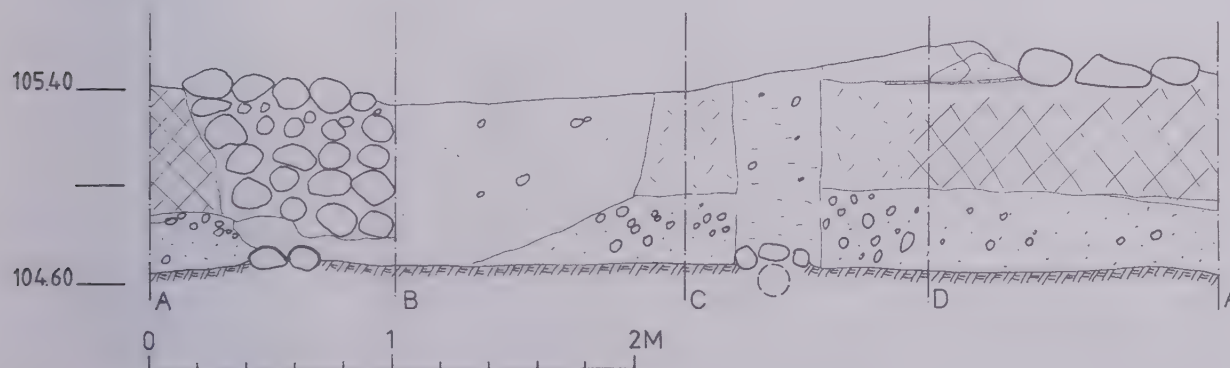
The underground water pipe which is known to extend along the entire southern boundary of the Royal Estate (from the Qelt Conduit to the area under



Ill. 73. Sounding AC78-A(C)144 with pipe-line at its bottom and Locus A(C)162 in background, viewed from the west. [E.N.]

discussion) was actually revealed by us across the full width of Area AC (see Plan 13). The following sections were exposed: ca. 4.5 m in Sounding AC77; ca. 1.6 m in Sounding AC78-A(C)144 (see Ill. 73); ca. 1.3 m in Sounding A(C)162-AC25 (see Ill. 74); ca. 9.5 m along the entire length of Structure AC14 (Loci AC10, AC14 and AC15); below the water-distribution installation in Locus AC1 and for 2 m to its east; in a small part of Sounding AC35. Traces of its existence were even found in Sounding AC53, the locality where the pipe was cut, most probably during the construction of the moat. We should mention that along the entire length of the pipe in Area AC, up to the water-distribution installation in Locus AC1, its course follows a straight line. Starting from the vicinity of Sounding AC35 eastward, the pipe turns slightly southward, toward the palace.

Along its entire length the underground pipe has a casing built of small fieldstones held together by lime-based plaster (containing much gravel). This casing, which provided protection and made possible the build-up of water pressure, completely enveloped the pipe (ca. 10–15 cm on three sides). It is important to mention that from its starting point (close to the northwestern corner of the Royal Estate, ca. 400 m to the west of Area AC) up to wall W1001, the pipe follows the line of the estate's southern wall (here wall W313); on the other hand, to the east of wall W1001 there were no traces of the estate's wall (apparently it turned northward here), but terrace wall W1069, undoubtedly intended, *inter alia*, to protect the pipe, was uncovered to its south. Also of importance is the interrelationship between the above-mentioned elements: in our view, the underground pipe, wall W1001 and terrace wall



Ill. 74. Sections through Sounding A(C)162-AC25 with the underground pipe. (For location sections, see Plan 13)

W1069 were built in conjunction, as was inferred from Soundings AC25 and AC22.

All along the section in which we had uncovered the underground pipe, we found no hint of any outlet for the water — in the form of a pipe, channel, etc. This raises problems with regard to the route along which water was supplied to Pools A(C)94 and AC44 (described below), before the Na'aran Conduit was built. Apparently we should assume that the outlet from the pipe was at a point to the west of the area excavated by us. (Alternatively, if this is not the case, the outlet may be hidden within wall W1101.)

The construction of the southern branch of the Na'aran Conduit (see Ills. 75, 76), in Stage 3, undoubtedly influenced the entire structure of the water system here. There were no problems with regard to its course. The conduit was built in an absolutely straight line dictated, in this sector too, by the southern wall of the Royal Estate. In all probability, the underground pipe had not yet gone out of use at this stage. One of the clearest signs of this is the fact that the small pool for the distribution of water (in Locus AC1) was situated on top of the pipe (see Ill. 77) and adapted to it by means of a 25-cm-high step located on its bottom. It is to be assumed that a supply of water under pressure by means of the pipe was necessary for the various installations within the palace. The new conduit,



Ill. 76. Na'aran Conduit within the boundaries of Area AC, viewed from the west. To the left of the conduit are remains of wall W1022. Note remains of a small channel on top of the conduit's floor, in foreground.



Ill. 75. Na'aran Conduit within the boundaries of Area AC, viewed from the west. Note the depression caused by the erosion into the moat. [E.N.]



Ill. 77. A small pool for distribution of water, Locus AC1, viewed from the south.

which was an open channel, could not provide such pressure.

The water-distribution installation in Locus AC1, which has been almost completely preserved, is a small pool (85×85 cm in area and 75 cm deep). The thickness of its walls was ca. 25 cm and it was completely coated with lime plaster. Water was fed into this installation via a 20-cm-wide channel which issued from the Na'aran Conduit, at the level of the



Ill. 78. Small channel connecting the Na'aran Conduit with Installation AC1.

latter's bottom (see Ill. 78). (It should be mentioned that the last section of this channel, 1.0 m in length, leading up to the installation has been destroyed.) The water flowed out of the distribution installation at the level of its rim, and thus the pool served also as a settling tank. From the pool the water was emptied into three channels:

1. a small channel which runs southward and has been preserved for only one meter (see Ill. 79);
2. a small channel which runs to the southeast and has been preserved over a length of ca. 4 m (see Ill. 80);²
3. a pipe emerging from the installation between the two channels and following a course parallel and close to the second one for a length of at least 4.0 m.

The water in the distribution pool was led into the



Ill. 79. Installation AC1 with channel leading south in foreground and channel leading southeast in background, viewed from the west. Note wall W1069 in front of and behind the installation.



Ill. 80. Installation AC1 and the outlets of the two small channels. Outlet pipe is hidden in the installation's wall between the two channels.

pipe by means of a funnel attached to the rim of the pool. For the construction of this funnel secondary use was made of a storage jar. The channel extending to the southeast and the pipe next to it were laid at the same time — the pipe's lime plaster casing forms one sidewall of this channel. The purpose of the first-mentioned channel was apparently to supply water to the two swimming pools, A(C)94 and AC44. On the other hand, the pipe was evidently intended for leading water to the area of the palace. [It is possible that the pipe we uncovered on the west of the Northern Tell, in Loci AA34, AA18 and AA16, is its continuation.] The second channel probably provided the palace and the nearby garden with an additional water supply. In any event, we located its continuation next to the above-mentioned section of the pipe in Locus AA34 (see Ill. 81). The intermediate section of the pipe as well as the channel was truncated by the moat.

The new conduit was also linked to Pool A(C)161, which was possibly built following its construction. Here too this was achieved by means of a small channel which led off from the southern side of the conduit at the level of its bottom.

Another water line laid in the sector under discussion, actually the sole water source of the Fortified Palace, is a pipe attached to the southern sidewall of the Na'aran Conduit. We became acquainted with this pipe at its start next to the meeting point of the Qelt Conduit and the discussed southern branch of the Na'aran Conduit (in the southwestern corner of the Royal Estate). For



Ill. 81. Channel and pipe, probably part of the water supply system to the Buried Palace, in Locus AA34, viewed from the north.

approximately half of its length the pipe is inserted into the southern sidewall of the conduit, and from there on, it is attached on the outside, within a wall specially built for this purpose (ca. 35 cm thick and 50 cm high).

Despite the fact that the link between this pipe and the pair of pipes exposed on the slope of the Northern Tell (in its northwestern sector) has been severed as the result of the moat's collapse, we have no doubts that such a connection did indeed exist. In any event, all of these pipes were part of the siphon system for raising water to the top of the artificial mound. In all likelihood, at the point where the individual pipe reached the moat there was an installation (evidently located behind the stone wall surrounding the moat) by means of which the water was lowered to the bottom of the moat and then raised to the top of the mound, with the aid of the pair of pipes.



Ill. 82. Lead pipe inserted into a ceramic pipe exposed in Locus AC20, viewed from the southwest.

It seems that although the main function of this siphon system was to supply water to the elevated Fortified Palace, it had at least one other purpose. An outlet from the pipe under discussion was exposed

below Structure AC14, at a point located ca. 9.0 m to the northwest of the distribution pool (AC1). Here we uncovered a short section of a lead pipe whose end was inserted into this ceramic pipe (see Ill. 82). We lack any data on what use was made of this water.

On the eastern margins of Area AC we exposed the beginning of the Small Bypass which replaced that section of the Na'aran Conduit which had been cut off during the construction of the moat (Stage 4). Although the point at which the bypass conduit branches off from the Na'aran Conduit has been damaged by the erosion of soil into the moat, as were considerable sections of its continuation, the course of the bypass conduit can nevertheless be clearly distinguished (see Ill. 83).

Also located on the northern periphery of Area AC, is the Grand Bypass, the conduit which branches off from the Na'aran Conduit in the direction of the Hasmonean Garden on the north of the Pools



Ill. 83. The Na'aran Conduit at the point where erosion into the moat caused its destruction, viewed from the east.

Complex. It begins at the northwestern corner of Area AC, in Sounding A(C)193. The location of its starting point and orientation of its course were apparently dictated by the presence of Pool A(C)90. We tend to date the construction of this conduit to Stage 5, although there is also the possibility that this took place during Stage 6 (but no later). Even after the Grand Bypass was completed, water apparently continued to flow through the above-mentioned section of the Naʿaran Conduit, though in smaller quantities. This water was evidently intended for Pool A(C)90, for the above-described distribution pool in Locus AC1, and for the Pools Complex [A(B)101] (see Chapter 3).

Structure AC14, though undoubtedly associated with Pool A(C)90, was probably built later, at the time when changes were made in the water system. It was erected above remains of the Naʿaran Conduit, which was partly dismantled, apparently at the same time. However, whereas most of the conduit's northern sidewall was now demolished (see below), all of its southern sidewall was left intact, probably to enable the ceramic pipe attached to it (see above) to continue to function.

How water was fed into the small distribution pool, AC1, following the severance of the Naʿaran Conduit, remains an open question. A 20-cm-wide channel, built on the bottom of the Naʿaran Conduit and with its starting point south of the southwestern corner of Pool A(C)90, was apparently meant to solve this problem. However, this channel, only 1.5 m of which have been preserved (see Ill. 76), is located within wall W1022. We do not know whether the small channel predated the wall and was destroyed during the latter's construction, or whether both were built at the same time. (The difficulties stem from the fact that although the small channel could not have been constructed prior to the destruction of the above-mentioned conduit's sidewall, a section of the northern side of the conduit was integrated into wall W1011, suggesting that the conduit was intact until this wall was built.)

In summarizing the discussion of the water system in Area AC, we should mention that even though certain details are not clear, we have a coherent general picture which conforms with the stage-wise development of the structures here and also in other parts of the palace complex.

THE PAIR OF SWIMMING POOLS

A(C)94 AND AC44

(and Their Surroundings)

At the center of Area AC are located two swimming pools which, in our opinion, form its focal element. We shall first describe the two pools and then the area surrounding them.

Pool A(C)94 (including Loci AC17, AC19 and AC21; see Ill. 84)

Pool A(C)94, 8.8 m long and 8.2 m wide, is flanked by walls 1.0 m thick, which have been almost totally preserved, particularly on the west and north. A 1.7-m-wide flight of stairs, almost completely preserved, descended from the southwestern corner, along the western wall, to the pool's bottom. Some of the steps have treads measuring 35 cm and others have larger ones (landings of a sort). The maximum depth of the pool, in the northeastern corner, was 3.7 m (only 3.3 m in the southwestern corner). In the northwestern corner and to some extent also in the southwestern corner, evidence of a bench built into the pool's wall has been preserved. It is clear that this bench originally framed the entire pool. Its width was 45 cm as was its height. Waterproof ash-lime plaster remained on most of the floor and walls. In any event, it was possible to distinguish two layers of plaster on the walls — the original one and another layer testifying to the renewal of the plaster over the years. [We should mention that the lowest two steps were built at a later (apparently technical) stage than the others, if we are to judge by the seam located here.]

Initially, water was fed into the pool by means of a ceramic pipe whose end was found to be built into the northern sidewall, close to the northwestern corner. The pipe approaches the pool diagonally from the northwest and has been preserved for a length of only 3.3 m. Beyond that point it was dismantled, in all likelihood at the time when construction works were carried out around the pool, perhaps including the erection of wall W316 (see below.) After this pipe went out of use, water was apparently supplied via a channel (15 cm wide) coming from the north and ending opposite the center of the pool's northern sidewall (see Ills. 72, 85). This channel, which



Ill. 84. Pool A(C)94, viewed from the northeast.

probably issued from the water installation in Locus AC1 (see above), is missing from a point ca. 3.5 m north of wall W316.

Pool AC44 (including Loci AC79, AC80 and AC81)

Only the western half of this pool has been preserved (see Ill. 86). The eastern half was completely demolished when the moat was dug around the Fortified Palace. The length of the pool was 8.8 m, being actually equal to that of Pool A(C)94; its assumed width, by analogy to Pool A(C)94, was 8 m.



Ill. 85. Area north of Pool A(C)94, viewed from the north. Room AC2 in center; southwest corner of Structure AC14 in foreground.



Ill. 86. Southwestern corner of Pool AC44. Note the section of the southern wall destroyed due to the construction of the moat.

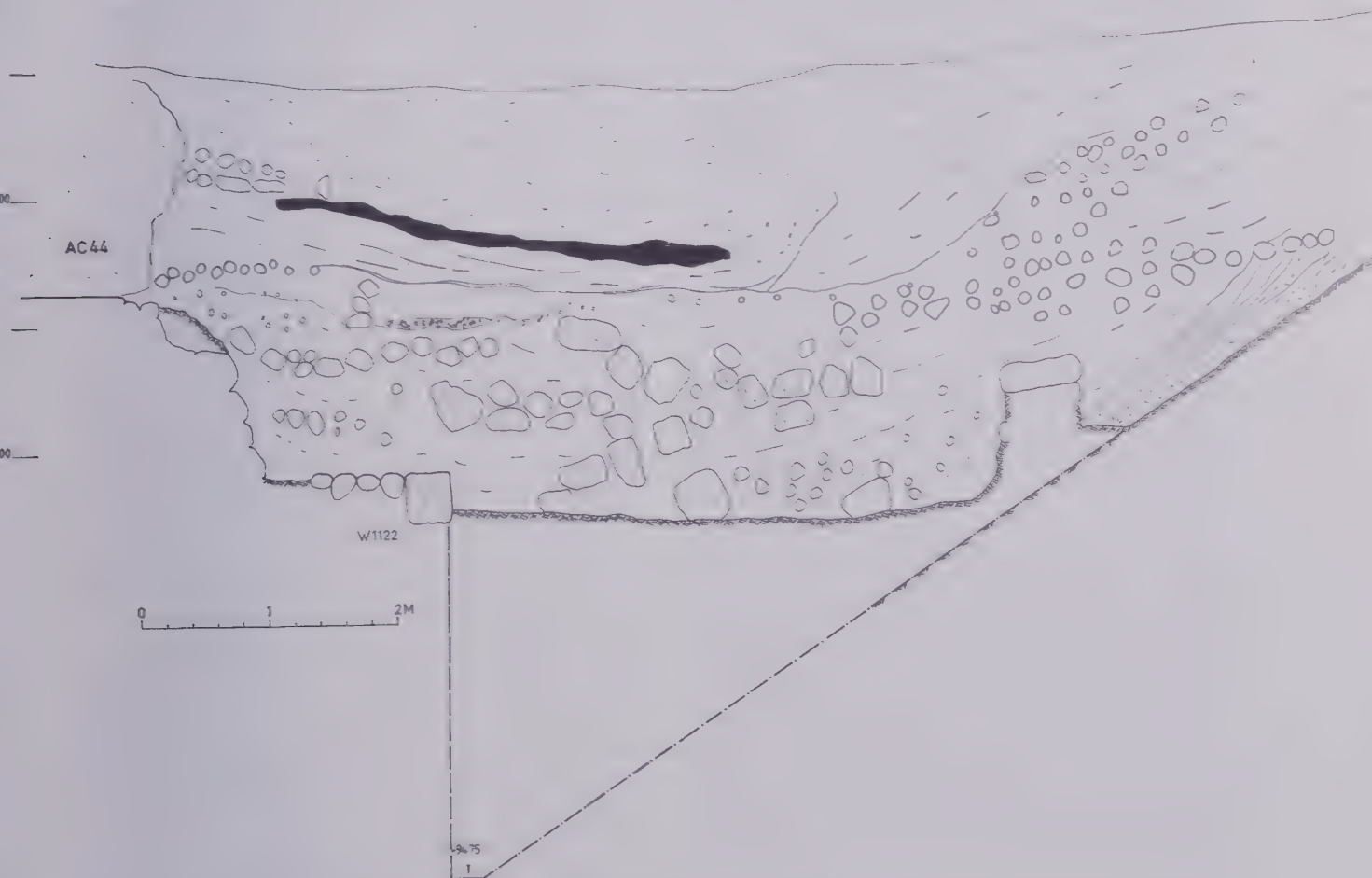
Its structure was also very similar to that of the other pool: walls with a thickness of 1.0 m and a flight of steps in the same position, along the western sidewall. [However, in this case the width of the flight was 1.8 m and not 1.7 m as in Pool A(C)94, and most of the treads of the steps measured 45 cm.] Only the upper parts of the sidewalls are missing (they were perhaps dismantled at the time when the pool went out of use) but one can assume that here too benches were built into them. The pool is coated with a single thick layer of ash-lime plaster which is excellently preserved.

We exposed but part of what remains of this pool. Its bottom (at level +99.50) was revealed only on the northern side, in AC49 at the western end of Trench AA11 (see Ill. 87). [The depth is similar to that of Pool A(C)94.] However, in contrast to the latter, whose bottom had been filled, in all likelihood gradually,

with silt and yielded many finds, in Pool AC44 the earth exposed was of uniform composition, similar to that of a deliberate fill, and contained only a few finds.

Unlike the situation in Pool A(C)94, here we did not obtain data relating to the supply of water. It can be assumed that both pools were fed in the same manner, i.e., originally (in Stage 2) by means of a pipe and later (in Stage 3) via a channel.

Pool AC44 was flanked on the north and west by a series of walls (two on each side), which were apparently built at the same time as the pool. The thickness of all the surviving walls (built of small fieldstones) was 75 cm. One of the two walls on the west, W318, was erected next to the pool's sidewall, and the other, W317, was located at a distance of 2.4 m from it. [It should be mentioned that the latter wall stood only 15 cm away from Pool A(C)94.] On the



Ill. 87. Section through Trench AC49 featuring the plastered sidewall of Pool AC44, viewed from the south. (For location section, see Plan 12)

north, wall W1084, of which only 1.0 m has survived, was built at a distance of 50 cm from the pool's sidewall, and wall W1085, of which 4 m have survived, at a distance of 2.1 m from it. Similar walls had possibly been located also to the east and south of the pool but were demolished either prior to or during the construction of the moat.

Water Systems to the South and Southwest of Pool A(C)94 [the area described here includes the following loci (from east to west): A(C)145, A(C)196, AC9, AC45, AC43, AC8, A(C)194, AC6, A(C)147]

A particularly intensive concentration of water installations was exposed by us to the south and southwest of Pool A(C)94. These installations are well preserved, despite their close proximity to the ground surface. However, they are not complete and therefore it was not always possible to comprehend their function. Actually preserved here were ceramic pipes, water channels and lead pipes (mainly their impressions).

One section of a ceramic pipe was revealed at the eastern edge of the area under discussion, next to wall W317; it extends diagonally, toward the northwest. Three segments of this pipe, each with a diameter of ca. 9 cm, were uncovered here, the first one penetrating the above-mentioned wall. This segment was L-shaped in order to receive a vertical pipe (built within the wall and probably serving to collect water drained from some roof; see below). Water in this pipe apparently flowed into Pool A(C)94.

Close to the above-mentioned ceramic pipe we found a small channel which began some 50 cm to the east of wall W317. The channel was plastered and in certain sections even its original cover slabs were preserved. It is most reasonable to assume that it issued from Pool AC44 and that its purpose was to drain off the overflow from this pool. This channel continues westward, taking a slight turn to the south, and was exposed for a length of ca. 20 m. (The elevation of the channel's bottom in the west is 37 cm lower than at its starting point in the east.) It is important to mention that at the meeting point of the channel and the ceramic pipe, the former passes over the latter; thus, from a technical viewpoint, the pipe

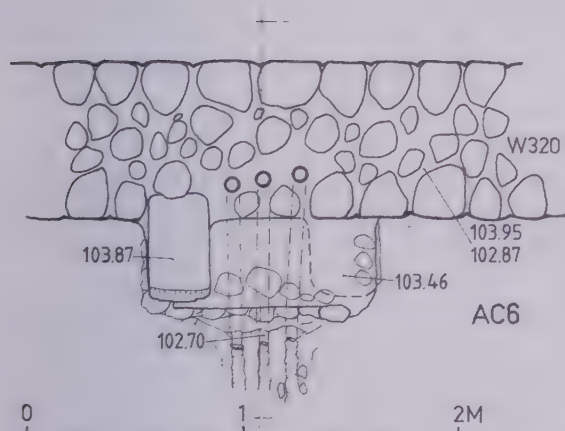
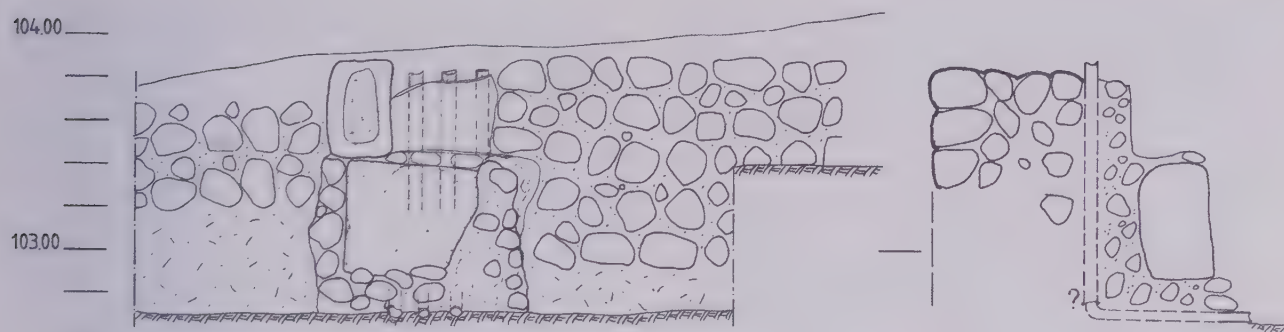


Ill. 88. A network of small plastered channels exposed immediately west of the southwestern corner of Pool A(C)94, viewed from the south.

preceded the channel. In the last 5–6 m on the western side, the channel broadens and becomes deeper. Beyond the end exposed here, which seems to be an intentional termination, the channel apparently continued not as a built element but dug into the ground.

Another network of built plastered small channels was exposed immediately to the west of the southwestern corner of Pool A(C)94 (see Ill. 88). The channels are located in the area between the pool and wall W319. There appear to be two channels here, one issuing from the pool itself and the other starting at its outer edge. These channels are linked to one another in an unusual way (in the form of the letter A). At the apex of the "letter" they join to form a single channel which passes through the foundation of wall W319. The direction of flow of the water was apparently toward the west, but the continuation of the channel has been destroyed and it is thus difficult to determine its exact course. [Theoretically, it is possible that the channel to the north of the pool, which drained Rooms AA2 and AA23 (see below), continued southward and linked up with the channel under discussion. Although we were unable to discern the entry of the hypothetical channel into the above-described one, the linkup is possibly concealed below a balk which has not yet been excavated.]

The last and most unusual system consists of three lead pipes discovered here (see Ill. 89). It begins in wall W320 which has a northwest orientation and stands 10 m to the west of Pool A(C)94. Here, within



III. 89. Detailed plan of system of water pipes in Trench AC6.
Top left: elevation of the installation, viewed from the east.
Top right: section through the installation, viewed from the south.

wall W320, are located three ceramic pipes of relatively small diameter (ca. 7 cm each), which descend vertically for about 1 m. The pipes are actually situated within a sort of pilaster (length ca. 100 cm, width ca. 35 cm) abutting the wall (see III. 90). At the base of the pilaster each of the ceramic pipes is joined to a lead pipe, of which only the end has survived since all the other parts were looted in antiquity. (Incidentally, the end of each of the lead pipes is distended as the result of forceful looting operations.) From here onward these pipes ran parallel, one next to the other, to the east, ca. 30 cm below the original floor which has disappeared. The pipes themselves are missing but have left clear impressions since they were laid within a substratum of ash-lime mortar characteristic of water



III. 90. Wall W320 in background with evidence of three lead pipes in front of it, viewed from the east. Note pilaster housing three vertical ceramic pipes.

installations. Some 6 m beyond wall W319, all traces of the southernmost pipe vanish, and the two remaining ones turn to the southeast. They continue side-by-side for another 6 m and then part company. The northern pipe continues eastward in a straight line for a further 5 m and then also disappears, perhaps not by chance in front of two stone slabs, 35×70 cm in size, which we uncovered here. The other pipe (originally the middle one) continues for another 4 m and then all traces of it vanish next to a round depression (diameter 40 cm), coated with hydraulic ash-lime plaster, which was uncovered here, ca. 3 m to the west of the two stone slabs. One can only surmise the function of these lead pipes. Apparently they were connected to water installations serving as taps and/or fountains, but we have no proof of this. These three pipes apparently received water from the underground pipe which supplied water to the palace in Stage 2.

Wall W1001 and the Two Rooms AC2 and AC23 Adjacent to It

Wall W1001, which has a thickness of 1.1 m and extends in a north-south direction (with a slight bend toward the west), was exposed for a length of ca. 8 m in the area between the Naʿaran Conduit and Pool A(C)94. This wall, which actually forms a revetment to support the earth to its west (despite the fact that both of its faces are built like those of a regular wall), abutted the northern end of wall W1069 which runs from west to east. The latter was probably not a regular wall but a low revetment, a kind of garden



Ill. 91. Room AC2, viewed from the west. Wall W1001 in foreground.

terrace, intended to support and protect the underground water pipe laid next to it. (As mentioned above, this pipe supplied water to the palace during its early years.) Wall W1001 is therefore one of the earliest elements in Area AC. What is not clear is the situation at its southern extremity. Here the end of the wall, as exposed, appears to be truncated (see Ill. 91), and it is not clear how far it continued originally, and what was its connection with walls W1008 and W1009, to its west. Hypothetically, in the earliest stages of this area, wall W1001 could have been the western boundary wall of the area west of the Buried Palace. When this area was later extended westward in order to build Pool A(C)94, the southern section of wall W1001 was dismantled and wall W320 was built instead (see below).

Adjacent to wall W1001 on the east were two small rooms, AC2 and AC23 (see Ill. 85). Their common eastern wall W1004 is not parallel to wall W1001 but is aligned with Pools AC44 and A(C)94, as are wall W1005 which bounds Room AC2 on the south, and wall W1002 which separates the two rooms. (Incidentally, all three of the last-mentioned walls are 50 cm thick.) In this context, we should mention that the location of wall W1005 indicates that wall W1001 originally extended at least 50 cm beyond the stump that has remained (see above).

Rooms AC2 and AC23 were intended for the intensive use of water, as indicated by the following features:

1. Waterproof ash-lime plaster which coats the floors and walls of these completely preserved rooms. Most of the walls have survived but only to a height of a few centimeters above the floor. On the other hand, wall W1001 (and the plaster on it) has been preserved to a relatively great height;
2. Drainage of water from the rooms by means of a lead pipe which was uncovered *in situ* in the southeastern corner of Room AC2 (see Ill. 92). [This pipe is connected to a stone-lined channel, coming from the area east of the two rooms, and continuing westward beyond the corner formed by walls W316 and W319, after which it disappears (see Ill. 93)].

We assume that the two rooms under discussion served as bathrooms and/or dressing rooms for all or



Ill. 92. Lead pipe, *in situ*, revealed in the southern wall W1005 of Room AC2.



Ill. 93. Wall W316 with the channel from Rooms AC2 and AC23 adjacent to it, viewed from the east. Note the two different building techniques in the wall behind. [E.N.]

some of those who used the two swimming pools, A(C)94 and AC44. At a later time, apparently during Stage 6, when walls W1006 and W1007 were erected, these two rooms went out of use.

Pool A(C)161: A square pool, with an area of 3.6×3.6 m and a depth of 3.9 m, was exposed to the west of revetment wall W1001 and to the south of wall W313, the boundary wall of the Royal Estate. The pool was partly uncovered and its bottom was reached only in a small section in the northeastern corner. The pool's sidewalls were 85 cm thick and were coated, on the inside and on top, with lime plaster. Apparently, these sidewalls were flat at the top, without any benches. With the exception of pools which served as the *otzar* of a *mikveh* and those which formed part of industrial installations (here in Jericho), this pool is perhaps the only one at the winter palaces' site that does not have a flight of steps descending to its bottom. We therefore regard it as a pool intended mainly for the storage of water.

Water was supplied to this pool, as mentioned above, by means of a small channel which issued from the Na'aran Conduit (via a ceramic pipe which was inserted into the latter's sidewall at the level of its bottom). Most of the channel has been preserved, including some of the stone slabs which covered it (see Ill. 73). The pool is indeed very close to the underground ceramic pipe built in order to supply water to the palace in Stage 2 (the distance between them is only ca. 35 cm). However, not only is the level of the pipe ca. 70 cm below the rim of the pool but no connection between them has been revealed. In any event, it seems that the builders of the pool (which, in our view, was constructed in Stage 3, following the erection of the Na'aran Conduit) were aware of the existence of the underground pipe, although it was hidden from sight, and built the pool next to it. The pool's orientation, in all events, was determined in accordance with wall W313 (the southern boundary of the Royal Estate), located only 1.3 m from its northern sidewall. On the other hand, the orientation of Pool A(C)161 differs from that of wall W1001. Therefore the pool is attached to this wall only in its southeastern corner, giving rise to a slight truncation of this corner. This interrelationship indicates that wall W1001 was built before the pool (see Ill. 94).

Another water installation — a very small pool —



Ill. 94. Trench AC25 with the eastern sidewall of Pool A(C)161 to the left, and wall W1001 topped by wall W314 of Structure AC14 to the right, viewed from the south. [E.N.]

was revealed near the northwestern corner of Pool A(C)161, to the immediate south of wall W313. Its area was 50×50 cm and its depth was 70 cm. The rim of this pool is ca. 30 cm above that of Pool A(C)161. There is no hint as to how water entered and was drained from it. Therefore it is difficult for us to assess its function, though its size indicates that it possibly served as a settling tank; however, as mentioned above, the system of pipes and/or channels which were connected with it are missing.

To the south of Pool A(C)161, two truncated east-west walls, W1008 and W1009, were revealed adjacent to one another. Wall W1008 forms a corner with wall W320, the western boundary wall of the area under discussion. Close to the corner, a channel, attached to wall W1008, was exposed. We do not know where this channel began or ended. (It was either eliminated by wall W1009, or, alternatively,

integrated in it.) Further to the east, and in line with the southern face of wall W1009, a section of white sloping plaster was revealed *in situ*, serving as proof that this wall continued to the east (see Ill. 95). These walls can be regarded as the terrace wall for the area to the north, containing Pool A(C)161, but it is not clear to which stage they belong.

Layout of the Area around Pools A(C)94 and AC44

One of the most difficult problems in Area AC is the interpretation of the structures which existed around the two pools, A(C)94 and AC44, mainly to the north and west. As was explained above, Pool AC44 was probably built a few years before Pool A(C)94. It appears that the former, when completed, was



Ill. 95. Walls W1009 (left) and W1008 (right), with a section of sloping plaster in line with wall W1009, viewed from the east.

surrounded by a circumvallation, next to which (at least on the west and north) were possibly porticoes. It is also reasonable to assume that the space between Pool AC44 and the Buried Palace, and also between it and Wadi Shaqq ad-Dabi, an area of ca. 35×35 m, served as a formal garden into which the pool was integrated. Although the pools under discussion do not have the same orientation as the palace, their alignment is generally similar. [If we were to extend wall W1085, located to the north of Pool AC44, in a straight line up to the palace, it would reach the northwestern corner of Room A(A)375.]

It seems that at this stage, development projects, mainly gardening, were carried out also to the west and north of Pool AC44. Walls W1069 and W1001, which had probably been built at the same time as the palace, formed part of the boundaries in the northern sector of this newly developed area. (This follows mainly from the close link between the ceramic pipe, wall W1069 and wall W1001.) As mentioned above, wall W1001 was found to be truncated, and might have formed the western boundary wall of the area under discussion in its earliest stages. When the area was extended westward in order to build Pool A(C)94, wall W1001 was mostly dismantled and wall W320 was erected to its west, forming the boundary of the area under discussion. Hypothetically, an east-

west wall connected the remaining northern section of wall W1001 with wall W320. This hypothetical link was later shifted northward, resulting in a northern extension of wall W320, as reflected by the vertical “seam”³ in it (see Ill. 96).

We lack absolute proof that Pool A(C)94 was built after Pool AC44. The great similarity and close proximity of the two pools theoretically would have called for better architectural coordination between them, if they had indeed been built at the same time (as was actually achieved in the Pools Complex in Area AB). Although the two pools are parallel to one another, their east-west sidewalls do not form a straight line despite the fact that they are of the same length (they are ca. 70 cm out of alignment). If both pools had been built together, we would also expect greater uniformity in the width and structure of the flights of steps, in the color of the plaster, etc. Even if this last argument is not fully convincing, it does add an element of probability. Wall W316, located to the north of Pool A(C)94, approaches and is partly even built upon the corner of walls W317 and W1085 which flank Pool AC44. Although the wall in question was built at a later stage than Pool A(C)94, this does hint at a general trend.

In any event, even if the leveling of the area around Pool A(C)94 had not been fully accomplished



Ill. 96. The vertical “seam” in wall W320, with different building techniques on either side of it, viewed from the east. [E.N.]

previously, it appears that it was completed up to wall W320 when this pool was built. On the south a 22-m-wide area appears to have been leveled up to a (hypothetical) western continuation of wall W235. At some stage several structures were built to the south of this wall (see below).

Within the limits of the area leveled around Pool A(C)94, both on the north and west, we exposed the walls of a single structure (at a distance of 3.3 m from the pool), which truly puzzled us [wall W316 to the north (see Ill. 93, to the left of the channel) and wall W319 to the west (see Ill. 97)]. Each of them has a leveled foundation 65 cm wide, built of a combination of fieldstones and hewn stone slabs. The stone slabs appear to be positioned in an orderly manner, in pairs, with a distance of ca. 1.7 m from center to center, but they do not continue regularly along the entire length of the walls, and it is therefore doubtful whether they hint at columns which stood above them, as appears to be the situation at a first glance. Above this leveled foundation we exposed remains of a wall (or balustrade) with an unusual structure: it is plastered on the side facing the pool but this plaster is

recessed ca. 20 cm from the edge of the foundation; at two points semicircular pilasters of sorts (actually shallow arcs less than a full semicircle, also fashioned from plaster) project from the plaster. In an attempt to understand the architectural layout, apparently after the construction of the moat, we suggest a colonnade flanking the pool at least on three sides. (However, the western wall, W319, was found to extend further southward than necessary for the hypothetical colonnade, and might indicate the existence of other structures to the south of the pool.) In this case, columns would have stood on the pool's sidewalls, apparently opposite the semicircular pilasters on the outer fringe of the colonnade. It should be mentioned that many fragments of colored plaster, including rounded ones which originated from the coating of columns, were uncovered in this area, though not *in situ*. The stack of fieldstones behind the line of plaster that has been preserved here (up to an average height of ca. 25 cm) could be *in situ* remains of a narrow wall or, alternatively, here was located a parapet of sorts (built of ashlar which were looted). According to the last assumption, the fieldstones exposed here were either inserted in their place, or fell into the empty space resulting from the intensive looting.

Comprehension of the area under discussion is hindered by the absence of any *in situ* floors in the vicinity of these two swimming pools. The numerous tesserae found here (both in groups or individually) led us to think that the floors which had existed here, around Pool A(C)94 and even around Pool AC44, were paved with mosaics which for some reason were dismantled in their entirety! In any event, both the architectural remains (either *in situ* or not in place) and the various sophisticated water installations uncovered around Pool A(C)94 hint at a quite magnificent architectural setup for which mosaic floors would have been fitting. However, to our regret, we are unable to suggest any "educated" reconstruction for this area.

Remains on the South of the Two Swimming Pools

Only isolated investigations were carried out by us on the south of Area AC beyond the above-mentioned strip south of Pool A(C)94. In three soundings (Loci



Ill. 97. Wall W319, west of Pool A(C)94, viewed from the south.

AC45, AC58 and AC47), 6–10 m south of the pool, remains of walls were revealed. In Sounding AC45 we found fragments of a wall, a tentative continuation of wall W319. We have no clue as to the function and date of either the rounded wall which was revealed in Sounding AC58 (a possible continuation of wall W317), as well as a section of an east-west wall exposed in Locus AC47. The latter wall, in any event, postdates the moat.

Area AC was apparently bounded on the south by the continuation of wall W235 which later served as the southern boundary of the moat. The area to the south of the latter wall was apparently occupied both by a garden area and a few structures. In any event, some 40 m to the southwest of Pool A(C)94 is located a small pool [A(C)150], 2.5×2.2 m in size, which was mentioned at the beginning of this chapter. This pool had been excavated before we started our work here. We dug a number of soundings close to it, in Loci A(C)154 and A(C)155, but failed to find any details of interest, aside from parts of walls which hinted at a building, evidently a small one, that was located here. Sounding A(C)156, ca. 20 m to the east of Pool A(C)150, revealed no building remains aside from a floor level at +97.41.

REMAINS ON THE NORTH OF AREA AC

The construction of the moat at the expense of the eastern part of Area AC (a building operation which brought in its wake, as mentioned above, also the partial destruction of Pool AC44) led to the development of this area's northern part, which apparently had been occupied previously by a formal garden if not by the periphery of the Royal Estate. In the light of the data obtained during the excavations, it is difficult to arrive at a definitive determination of the nature of Pool A(C)90 which is the dominant element in the northern part of Area AC. However, it appears that it too was basically intended for bathing and swimming and was built following the elimination of Pool AC44.

Pool A(C)90

Pool A(C)90 was discernible as a distinct depression even before we started excavating in this vicinity.

Our work in the area of the pool was minimal, being actually limited to four soundings mainly intended to determine the exact location of its four corners. In none of these soundings did we reach the bottom of the pool, but we were nevertheless able to establish the presence of a flight of stairs, evidently descending to the bottom, which starts in the southwestern corner and extends along the western sidewall. Pool A(C)90 shows great similarity to Pools A(C)94 and AC44 not only in this detail but also, and mainly, in its dimensions (length 8.8 m, width 7.7 m). (The width of the flight of stairs was 1.5 m.)

As already mentioned, it is difficult to refrain from linking the timing and reasons for construction of Pool A(C)90 with the abandonment of Pool AC44. According to this assumption, Pool A(C)90 was built at the same time as the Fortified Palace, during Stage 4. In any event, it is hard to explain the angular orientation of this pool in relation to the Na'aran Conduit and the two above-mentioned pools. The west-east axis of the pool is parallel to the Grand Bypass, which passes 3.3 m to its north. If the structures to the east of pool A(C)90 (see below) were built together with the pool itself, then this 'complex' might have dictated the orientation of the Grand Bypass, which, on the basis of various data, is considered to have been built in Stage 5. Alternatively, it is also possible that the pool and the structures to its east as well as the Grand Bypass were positioned parallel to some other structure or wall that bounded the estate here and is as yet unknown to us.

On being built, Pool A(C)90 was attached to the initial Na'aran Conduit, which was probably still in use at that time. (Since the pool is not parallel to the conduit, their only point of contact is at the pool's southwestern corner, while the southeastern corner is at a distance of 1.7 m from the conduit.) Water apparently was led to the pool through a channel, issuing from the Na'aran Conduit, at the pool's southwestern corner. During Stage 5, water apparently still flowed in the nearby section of the Na'aran Conduit and continued via the Small Bypass to the Pools Complex. In Stage 6, when the section of the Na'aran Conduit passing through the area of the Pools Complex was abandoned and construction of the Grand Bypass was completed, the northern sidewall of the Na'aran Conduit in this section was

apparently dismantled (evidently in order to make use of the fieldstones of which it was built) and Structure AC14 was erected (see below).⁴

Structure AC14

It appears that in Stage 6 or Stage 7, a rectangular structure (AC14), measuring 11.0×9.0 m, was erected parallel to Pool A(C)90 and above the remains of the Naʿaran Conduit (see above). All that is left of this structure are the foundations of its four outer walls: W314 on the west, W1006 on the south, W1011 on the east (found to be mostly destroyed), and W1022 on the north (see Ill. 98). The width of the three first-mentioned walls was ca. 1.1 m, while the last one was only 80 cm wide.



Ill. 98. The southeastern corner of Structure AC14, viewed from the east. Visible in the background is Distribution Pool AC1.

There is no doubt that the floor of Structure AC14 must have been laid on top of considerable fill; its assumed elevation was likely to have been +105.25 (and perhaps even higher). In any event, this assumed floor level is higher than the remains of the Naʿaran Conduit's walls found here within the limits of the rectangular structure (see above). It should be mentioned that 1.1-m-thick wall W1007, attached to Structure AC14 on the south, probably served as the foundation of a flight of stairs making possible the ascent to this structure. In Locus AC4, a section of a unique kind of decorated plaster, with a sort of a zigzag on its surface, was revealed (see Ill. 99). The original location of this plaster is unclear.



Ill. 99. Locus AC4 with sections of sloping plaster with a zigzag decoration, viewed from the south.

We have no clear assessment of the function of the rectangular structure. One of the more acceptable possibilities is that the surviving walls served as the foundations of a kind of pavilion, intended for the use of bathers and swimmers in Pool A(C)90, if the latter was indeed used for such purposes, a fact which we were unable to determine with certainty. If this supposition is correct, it is possible that the northern wall of the structure opened onto the pool via a portico. This would explain why it was thinner than the other walls.

One problematic aspect is the stratigraphic relationship between Structure AC14 and the pool to which it was attached. It seems more reasonable to consider that they were built at the same time. This is possible only if we assume that Pool A(C)90 was constructed in Stage 6. The only apparent difficulty here is the time gap between the abandonment of Pool AC44 and the building of Pool A(C)90. At the present stage of research this remains an open question, but in any event it has no special significance for comprehension of the development of the entire Hasmonean winter palace complex.

Structures to the East of Pool A(C)90

To the east of Pool A(C)90, in a sector ca. 25 m long and ca. 15 m wide, we exposed the remains of some structures. Their orientation is the same as that of Pool A(C)90 and Structure AC14 to its south, as well

as that of the Grand Bypass located 3 m to the north. These structures were undoubtedly damaged over the course of years by erosion into the moat. One can actually speak of two rooms, one measuring 4.6×4.0 m (Loci AC31, AC30, AC29 and AC27), and the other to its south, measuring 4.0×2.6 m (Loci AC69 and AC39). A third room existed to the south of the last-mentioned room (Locus AC41) but most of it seems to have been destroyed. The rooms had beaten earth floors and no specific finds were revealed in them which could indicate their function. Their walls, built of fieldstones with ashlar in secondary use fitted between them, were partly 80 cm thick and partly 100 cm thick. It is possible that to the east of the three rooms there was a courtyard, of which only the western wall W1070 and the northern wall W1013 have survived.

We have no explanation for the series of walls found to the west and north of the above-described rooms and courtyard. First we should mention the pair of parallel walls W1074 and W1075 (Loci AC71, AC32, AC68 and AC33) which begin at the eastern sidewall of Pool A(C)90. Each is 55 cm thick and they are 1.4 m apart. In our view, these walls played some role in a garden to the south, e.g., as the foundation of colonnades. It is difficult to regard them as part of a

building in view of both their dimensions and the link between them and Pool A(C)90. While wall W1075 is attached to the northwestern corner of the structure to the east (see above), wall W1074 continues eastward, but now as a broad wall, actually a kind of platform attached to wall W1013 (Loci AC40, AC12 and AC27). The area to the south of these two parallel walls was apparently occupied by a garden. The floor levels revealed in the structures to the east are ca. +104.00, while the assumed level of the pool's rim is +105.40. It seems that not only did the pool's eastern wall function as a retaining wall but there was also no direct connection between the above-mentioned structures and the pool to their west.

Remains to the West of Pool A(C)90

We carried out only isolated investigations to the west of Pool A(C)90. In Locus A(C)193 we located the point where the Grand Bypass emerges from the Na'aran Conduit.

In another sounding (AC42), ca. 4 m west of Pool A(C)90, no remains of structures were found, but two large boulders located here could have been part of the garden layout which apparently flanked the pool on the west, north and east.

CHAPTER 3

AREA AB — THE POOLS COMPLEX

Area AB is the largest of the various areas excavated in Zone A (see Plan 2). Its maximum length (from north to south) is ca. 140 m and its maximum width is ca. 100 m. On the west, Area AB borders partly on the Northern Tell (Area AA), partly on Area AC and partly on the Royal Estate; on the south, on Area AE (the Twin Palaces); on the east, partly on Area AG and partly on the Royal Estate; and on the north, on the Royal Estate (see Ill. 100). Although this area is mainly flat, with a slight slope to the north, in its southern part the ground surface drops steeply to the south, toward the Twin Palaces. Even prior to excavation, the following two elements were distinguishable within this generally level surface:¹

1. a large depression, partly flanked by stone walls (see Ill. 101), which was defined by Kelso and Baramki as a reservoir but subsequently identified as a pair of swimming pools [Locus A(B)101] which were later combined into one, larger pool;²
2. a hummock to the south of the depression, around which it was possible to discern stone walls forming a rectangle.³

The key feature of the extensive area under discussion here is the “Pools Complex” built in Stage 3 (see



Ill. 100. General view of Zone A from the north.



Courtyard B64 in the Third Palace, viewed from the north.



Excavations of the garden in Courtyard B64. [E.N.]



Frescoes revealed in the southwestern corner of Room B88. [Photograph taken by E.N. during conservation work by the Israel Museum team.]



Detail of the fresco of Room B88.



Fallen architectural elements as exposed in Courtyard B55.



The Southern Tell as viewed from the eastern colonnade of Courtyard B55.



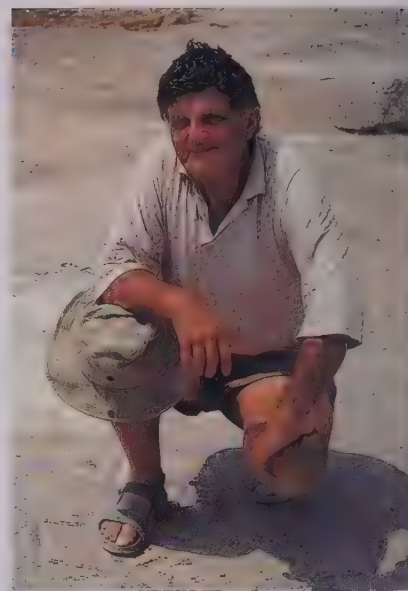
Herodian flowerpots exposed in various locations at the site.



Restoration work carried out in Courtyard B55.



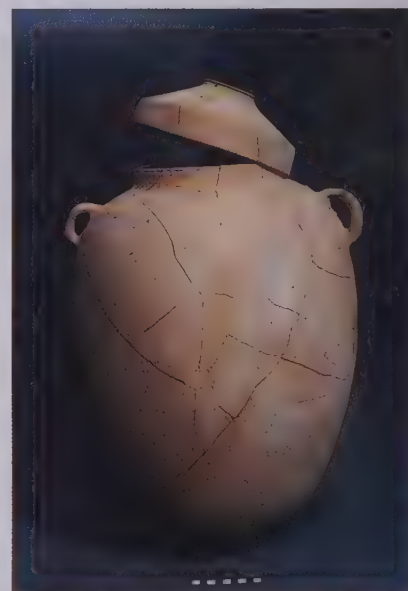
Ongoing work in Area AM with Nuseib Uweishira (to the north of Wadi Qelt) in the background.



Director Ehud Netzer.



A group of volunteers at the site, winter 1979. [E.N.]



A Herodian storage jar exposed at the site (similar to the ones found at Qumran), apparently used to store documents.



A Bedouin worker drinking from a Hasmonean bowl.



Breakfast -- Bedouin workers and team members eating together. [E.N.]



The expedition's headquarters. [E.N.]



Hoard of Hasmonean coins (all minted by Mattathias Antigonus) exposed in the Hasmonean Pools Complex.



Intact Herodian ceramic vessels exposed on the Southern Tell (Herod's Third Palace).



Ornate oil-lamp chandelier with 17 nozzles, exposed in the Roman Villa.



A volunteer at work. [E.N.]



A Bedouin gathering at the site (team member Egon Lass in foreground). [E.N.]



General view of the Plain of Temcho with Cyprus in the background to the left. (E. N. J.)



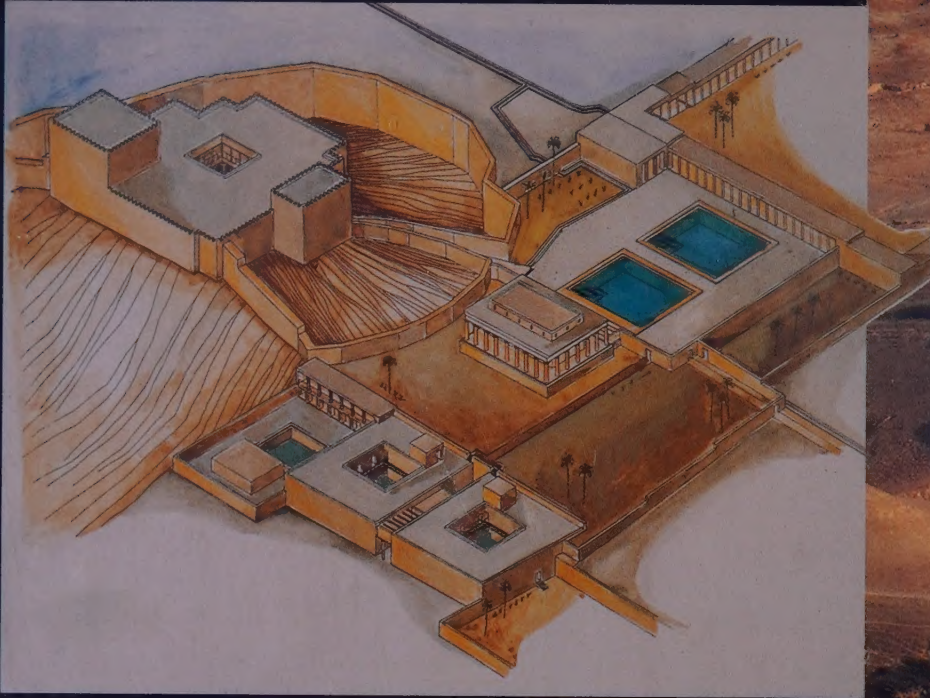
A luxuriant date palm grove in the area of the Elammanou Eloridian Royal Estate. (E. N. J.)

*Cover photograph:
General view of the Western Plain of Jericho with
remains of the Hasmonean and Herodian palaces,
viewed from the south
(photograph: Albatross, Tel Aviv).*

*Back cover drawing:
Isometric view of the reconstructed Hasmonean
Palace Complex at its zenith, viewed from the
southeast (concept by Ehud Netzer and drawing
by Jorge Salzberg).*

Hiericus (Jericho) is a plain surrounded by a kind of mountainous country, which, in a way, slopes towards it like a theatre. Here is the Phoenicon, which is mixed also with other kinds of cultivated and fruitful trees, though it consists mostly of palm trees: it is one hundred stadia in length, and is everywhere watered with streams and full of dwellings. Here are also the palace and the balsam park.

Strabo, *Geography*, 16.2.41



The oasis of Jericho was an agricultural center, a garden city and a winter resort for the Jerusalemite kings and aristocracy in the days of the Second Temple.

The Hasmonean kings had turned the western plain of Jericho into a huge royal estate where palm trees and balsam bushes were cultivated. Next to the estate, on the banks of Wadi Qelt, these kings erected a winter palace complex with numerous gardens and swimming pools. Herod the Great made this site (known as Tulul Abu el-'Alayiq) into a palatial center including three wings built over a number of years. The last one to be built was the "Third Palace," also known as the "Opus Reticulatum Palace," an outstanding architectural creation built on both sides of the wadi.

This volume, the first one in a series, is dedicated to the stratigraphy and architecture of the Hasmonean and the Herodian palaces, exposed and studied by the present author. Prof. Ehud Netzer, an architect and an archaeologist, has directed excavations at other sites as well, such as Lower Herodium, Caesarea Maritima, Masada, Sepphoris, Banias, and Jerusalem.

